

Production Supporting Systems in Factories

ระบบสนับสนุนการผลิตในโรงงานอุตสาหกรรม

Cloud Database

InfluxDB

- InfluxDB is a time series database (TSDB) developed by the company InfluxData.
- It is used for storage and retrieval of time series data in fields such as operations monitoring, application metrics, Internet of Things sensor data, and real-time analytics.
- Ranking
- Performance

Get Started with InfluxDB

- Register for an account at [InfluxDB Cloud](#).
- Create a new InfluxDB instance.
- Create a new organization and a new bucket.
- Create a new API token with all access permissions.

You will need

```
CLUSTER_URL=  
ORGANIZATION_NAME=  
BUCKET_NAME=  
API_TOKEN=
```

InfluxDB Data Model

- **Organization:** A container for all your InfluxDB resources.
- **Bucket:** A named location where time series data is stored.
- **Measurement:** A container for time series data, similar to a table in relational databases.
- **Tag:** A key-value pair that is used to store metadata about a measurement.
- **Field:** A key-value pair that stores the actual data points in a measurement.

Query Languages

- InfluxDB supports two query languages: `InfluxQL` and `Flux`.
- `InfluxQL` is similar to `SQL` and is used for querying time series data.
 - You will see this in Data Explorer in InfluxDB Cloud.
- `Flux` is a more powerful and flexible query language that allows for complex data processing and analysis.
 - You will use this in Node-RED to query data from InfluxDB.

Example of InfluxQL Query:

```
SELECT mean("value") FROM "your_measurement_name" WHERE time > now() - 1h GROUP BY time(1m)
```

Example of Flux Query:

```
from(bucket: "your_bucket_name")
  |> range(start: -1h)
  |> filter(fn: (r) => r._measurement == "your_measurement_name")
  |> group(columns: ["_field"])
  |> mean()
```

InfluxDB Node-RED Integration

- Use the `node-red-contrib-influxdb` package to integrate InfluxDB with Node-RED.